

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
Student Name:	G.Tejasree	Reg. No.:	192511204
Section / Batch:	CSE	Date:	10-08-2026

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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Answer: Grey level transformation is an important technique used in image enhancement. It changes the intensity or grey level values of pixels in an image according to a specific transformation function. The main purpose is to improve the visual quality of an image and make important details easier to identify.

It is mainly used to:

- Improve the contrast of an image.
- Make dark regions brighter or bright regions darker.
- Highlight important features and details in an image.
- Reduce the effect of poor illumination.
- Make an image more suitable for further processing and analysis.

Thus, grey level transformation helps to obtain a clearer and more useful image by modifying its pixel intensity values.

(b) Apply a negative transformation $s = 255 - r$ to the given pixel values:

$$r = [50, 100, 150, 200]$$

Given formula:

$$s = 255 - r$$

Calculation:

$$\text{For } r = 50: \quad s = 255 - 50 = 205$$

$$\text{For } r = 100: \quad s = 255 - 100 = 155$$

$$\text{For } r = 150: \quad s = 255 - 150 = 105$$

$$\text{For } r = 200: \quad s = 255 - 200 = 55$$

Therefore, the transformed pixel values are:

$$s = [205, 155, 105, 55]$$

~~Conclusion:~~ In negative transformation, the dark pixels become bright and the bright pixels become dark. For an 8-bit image, each pixel value r is replaced by $255 - r$.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Answer: Log transformation is an image enhancement technique used to expand the range of low-intensity pixel values and compress high-intensity pixel values. It is represented by the formula $s = c \log(1 + r)$.

How it enhances low-intensity pixels:

- It gives greater changes to small pixel values.
- Dark areas of an image become brighter.
- Details present in dark regions become easier to see.
- At high pixel values, the increase becomes smaller, so bright regions are compressed.

Therefore, log transformation is especially useful when important image details are hidden in dark or low-intensity regions.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

Given formula:

$$s = 10 \log(1 + r)$$

Calculation (using log base 10):

$$\text{For } r = 0: s = 10 \log_{10}(1 + 0) = 0.000$$

$$\text{For } r = 10: s = 10 \log_{10}(1 + 10) = 10.414$$

$$\text{For } r = 50: s = 10 \log_{10}(1 + 50) = 17.076$$

$$\text{For } r = 100: s = 10 \log_{10}(1 + 100) = 20.043$$

Therefore, the transformed values are approximately:

$$s = [0, 10.414, 17.071, 20.043]$$

Conclusion:

The log transformation increases the visibility of low-intensity pixels while compressing high-intensity values, thereby improving details in dark regions.

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Answer: Histogram equalization is an image enhancement technique used to improve the contrast of an image. It redistributes the pixel intensity values so that they are spread over a wider range of intensity levels.

Main roles of histogram equalization:

- Enhances the overall contrast of an image.
- Makes details in dark or bright regions more visible.
- Spreads frequently occurring intensity values over a wider range.
- Improves the visual quality of the image without changing its basic content.

(b) Given a 3-bit image with histogram:

Intensity:	0	1	2
Frequency:	2	2	4

Total number of pixels, $N = 2 + 2 + 4 = 8$

Step 1: Compute the cumulative frequency (CDF):

Intensity 0: $CDF = 2$

Intensity 1: $CDF = 2 + 2 = 4$

Intensity 2: $CDF = 2 + 2 + 4 = 8$

Therefore, $CDF = [2, 4, 8]$

Step 2: Compute equalized intensity values:

For a 3-bit image, $L = 2^3 = 8$, so $L - 1 = 7$.

Equalized value $s = \text{round}[(L - 1) \times CDF / N]$

For intensity 0: $s = \text{round}(7 \times 2/8) = \text{round}(1.75) = 2$

For intensity 1: $s = \text{round}(7 \times 4/8) = \text{round}(3.50) = 4$

For intensity 2: $s = \text{round}(7 \times 8/8) = \text{round}(7) = 7$

Final Answer:

Cumulative distribution: $[2, 4, 8]$

Equalized intensity mapping: $0 \rightarrow 2, 1 \rightarrow 4, 2 \rightarrow 7$

Conclusion:

Histogram equalization improves contrast by redistributing the available intensity levels. Here, the original intensities 0, 1 and 2 are mapped to 2, 4 and 7 respectively.